Performance fees can get complicated. If the HF has a *high-water mark*, then performance fees are levied only when the NAV exceeds the highest NAV previously achieved. For example, suppose the NAVs of a HF are:

Time	**1**	**2**	**3**	**4**
NAV	100	120	110	145

There is no performance fee payable at time 3 because the NAV is below its high-water mark of 120. At time 4, the HF recoups its losses and has a NAV of 145. A performance fee is paid at time 4 and the fund's water-mark is now 145. If there is a *hurdle rate*, say 4%, the performance fee is paid only when returns are above 4%. With both a high-water mark and a hurdle rate, the incentive fee is paid only when both conditions are met.

While the typical combination of management and performance fees is "2 and 20" (2/20), there is some variation. The Medallion Fund of Renaissance Technologies, founded by James Simons, is one of the best performing HFs and may be the best HF in history. It charges 5/44. (No, you can't invest; it is *closed*.) The mean management and performance fees are 1.4/16.0 across all funds. Smaller firms generally have lower expense ratios and higher incentive fees (and remember, small HFs also tend to have better performance).[39]

Finally, HFs sometimes charge *withdrawal fees* when funds are redeemed.

Incentive Alignments

Do performance fees lead to better performance? Yes. Ackermann, McEnally, and Ravenscraft (1999) report that moving from a HF with no incentive fee to a HF with a 20% incentive fee increases the Sharpe ratio of a fund by 0.15, all else equal. Agarwal, Daniel, and Naik (2009) find that, consistent with agency theory, funds with better managerial incentives have better performance. In particular, the more the HF manager gets paid when the HF does well (the *performance sensitivity* of the incentive fee, in technical jargon), the better the performance of the HF. HFs with high-water marks and whose managers have larger equity stakes also tend to perform better.

On the other hand, asset owners should be wary of HFs increasing their fees due to past success. Agarwal and Ray (2012) show that HFs change their fees quite often, but fee increases are followed by *worse* performance. Agarwal and Ray find no change in (often subpar) performance when HFs lower their fees.

The lack of disclosure and the lack of control hurt investors. Lack (2012) tells colorful stories of how hedge fund returns are gamed (see Section 3.1) and how the lack of investor control encourages managers to prolong losing investments to maximize fees. Fraud thrives on opacity, and asset owners have to be careful.

[39] See Deuskar et al. (2011).